Economics H Review Guide

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Range: Before midterm 30%; M3 35%; M4 35%

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

Post-Midterm Content

Monetary Policy

Without money, trade would require **barter**: the exchange of one good or service for another

Every transaction would require a **double coincidence of wants**—the unlikely occurrence that two people each have a good the other wants

Most would spend a lot of time searching for others to trade with (a huge waste of resources)→unnecessary with **money**

The 3 functions of money

- **Medium of exchange:** an item buyers give to sellers when they want to purchase g&s
- **Unit of account**: the yardstick people use to post prices and record debts
- **Store of value:** an item people can use to transfer purchasing power from the present to the future

The 2 Types of Money

- **Commodity money**: takes the form of a commodity with intrinsic value
  - Ex: gold coins, cigarettes in POW camps
- **Fiat money**: money without intrinsic value, used as money because of govt. decree
  - Ex: US dollar bill

The Money Supply (the money stock)

- Def: The quantity of money available in the economy

- **Currency**: the paper bills and coins in the hands of the (non-bank) republic
- **Demand deposits**: balances in bank accounts that depositors can access on demand by writing a check

M1: currency, demand deposits, traveler's checks, and other checkable deposits

M2: everything in M1 plus savings deposits, small time deposits, money market mutual funds, and a few minor categories

Central Bank and Monetary Policy

- **Central bank**: an institution that oversees the banking system and regulates the money supply
  - 2 main goals*: keep inflation within target, maximize employment*
  - Regulators of commercial banks: supervise and regulate commercial banks to ensure financial stability, prevent risky behavior, and protect depositors
  - Bankers to governments: manage the government's accounts, help issue debt (e.g. bonds), and sometimes finance public spending indirectly
  - Bank's bank (the lender of last resort to banks): provide emergency liquidity to commercial banks to prevent bank runs and systemic collapse
- **Monetary Policy:** the setting of the money supply by policymakers in the central bank
- **Federal Reserve (Fed):** the central bank of the US

The Structure of the Fed

- Board of Governors (7 members) in Washington, DC
- 12 regional Fed banks located around the US
- Federal Open Market Committee (FOMC)
  - Includes the Board of Govs and presidents of some of the regional Fed Banks, decides monetary policy

Bank Reserves

- In a fractional reserve banking system, banks keep a fraction of deposits as reserves and use the rest to make loans
- The Fed establishes reserve requirements, regulations on the minimum amount of reserves that banks must hold against deposits
- Banks may hold more than this minimum amount if they want
- **The reserve ratio, R**
  - Fraction of deposits that banks hold as reserves
  - Total reserves as a percentage of total deposits

Bank T-account:

- T-account: a simplified accounting statement that shows a bank's assets and liabilities
- Banks' liabilities include deposits; assets include loans and reserves

The Fed's 3 tools of monetary control

1. **Open-market Operations (OMOs):**
   the purchase and sale of US government bonds by the Fed

- To increase money supply: Fed buys govt bonds, paying with new dollars.
  - Which are deposited in banks, increasing reserves
  - Which banks use to make loans, causing the money supply to expands
- To reduce money supply: Fed sells govt bonds, taking dollars out of circulation, an the process works in reverse
- OMOs are easy to conduct, and are the Fed's monetary policy tool of choice

2. **Reserve Requirements (RR):**
   Affect how much money banks can create by making loans

- To increase money supply, fed reduces RR
  - Banks make more loans from each dollar of reserves, which increases money multiplier and money supply
- To reduce money supply, Fed raises RR
  - And the proccess works in reverse
- Fed rarely uses reserve requirements to control money supply: Frequent changes disrupt banking

3. **The discount rate:**
   the interest rate on loans the Fed makes to banks

- When banks are running low on reeserves they may borrow reserves from the Fed
- To increase money supply, Fed can lower discount rate, which encourages banks to borrow more reserves from Fed
- Banks can then make more loans, which increases the money suppply
- To reduce money supply, Fed can raise discount rate
- If no crisis, Fed rarely uses discount lending—Fed is a "lender of last resort"

The Federal Funds Rate

- On any given day, banks with insufficient reserves can borrow from banks with excess reserves
- The interest rate on these loans is the Federal Funds Rate
- The FOMC uses OMOs to target the Fed Funds Rate
- Many interest rates are highly correlated so changes in the fed funds rate cause changes in other rates and have a big impact

CHECK PPT: Monetary Policy and the Fed Funds Rate

- To raise fed funds rate, Fed sells govt bonds (OMO)
- This removes reserves from the banking system reduces supply of federal funds
- →causes r_f to rise
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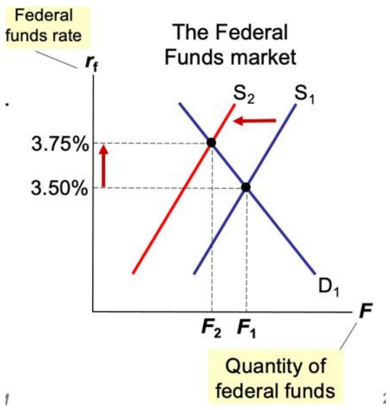
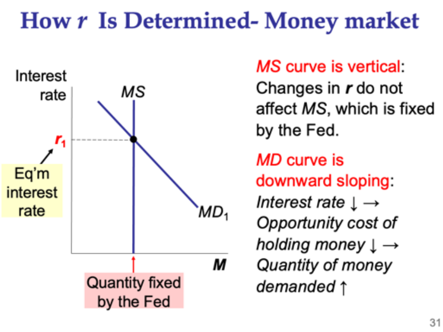


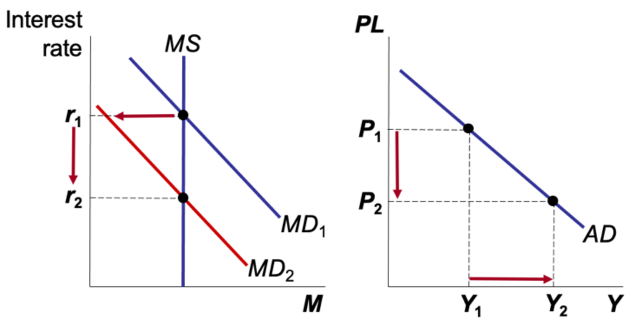
The Theory of Liquidity Preference

- A simple theory of the interest rate (denoted r)
- R adjusts to balance supply and demand for money
- <u>Money supply</u>: assume fixed by central bank does not depend on interest rate
    - =currency (cash)+ demand deposit
- <u>Money demand</u> reflects how much wealth people want to hold in liquid form
- Suppose household wealth includes only two assets:
    - Money—liquid but pays no interest
    - Bonds—pay interest but not as liquid
- A household's "money demand" reflects its preference for liquidity
- The variables that influence money demand Y, r and P

Money Demand: suppose real income (Y) rises.

> Households want to buy more g&s so they need more money
> To get this money they attempt to sell some of their bonds
> i.e. An increase in Y causes an increase in money demand, other things equal





Example: recessionary gap→lower FR + lower RRR + buy govt bonds→increase money supply→decrease interest rate→investment increase→AD increase (shift left)
Inflationary gap→raise FR + raise RRR + sell govt bonds→decrease money supply→increase interest rate→investment decrease→AD decrease (shift left)
Monetary expansionary policy only shifts AD

MS increase, IR decreases→ I increase (cost of borrowing decrease), C increases (savings decreases), EX increase (currency depreciates and cheaper exports)

Fiscal Policy

Fiscal Policy: the setting of the level of govt spending and taxation by govt policymakers

Expansionary fiscal Policy:

- an increase in G and/or decrease in T
- shifts AD right

Contractionary fiscal policy

- a decrease in G and/or increase in T
- shifts AD left

Fiscal policy has two effects on AD

1. **The Multiplier Effect**: the additional shifts in AD that result when fiscal policy increases income and thereby increases consumer spending

Marginal Propensity to consume

- That fraction of extra income that households consume rather than save
- E.g. if MPC= 0.8 and income rises $100, **c** rises $80→MPC= $\Delta C / \Delta Income$

MPC+MPS=1

$\Delta C + \Delta S / \Delta \text{income} = 1$

$$\Delta Y = \boxed{\frac{1}{1 - MPC}} \Delta G$$

The multiplier

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The Multiplier effect: each \$1 increase in G can generate more than a \$1 increase in agg demand; also true for the other components of GDP

2. The Crowding-out Effect

- Fiscal expansion raises r (G↑or I ↑, AD ↑, PL ↑(MD curve rightwards), R ↑), which reduces investment, reducing the net increase in agg demand.
- So, the size of the AD shift may be smaller than the initial fiscal expansion

Changes in Taxes

- A tax cut increases house holds' take-home pay
- Households respond by spending a portion of this extra income, shifting AD to the right
- The size of the shift is affected by the multiplier and crowding-out effect
- *Another factor*: whether households perceive the tax cut to be temporary or permanent
  - A permanent tax cut causes a bigger increase in C—and a bigger shift in the AD curve than a temporary tax cut

Fiscal Policy and Aggregate Supply

- Most economists believe short run effects of fiscal policy mainly work through AD
- A cut in tax rate gives workers incentive to work more, so it might increase the quantity of g and s supplied and shift AS to the right
- People who believe this effect is large are called "supply-siders"
- Govt purchases might affect AS
  - Ex: Govt increases spending on roads
  - Better roads may increase business productivity which increases the quantity of g&s supplied, shifts AS to the right
  - This effect is probably more relevant in the long run: it takes time to build the new roads and put them into use.

Budget deficit, budget surplus and debt

- Budget balance= tax revenue=Govt spending
- If the budget is in surplus in a financial year, the government is able to repay money that it has borrowed in previous years
- The national debt is the accumulated total of past government borrowing. It is the stock of government debt that is outstanding at a point in time

- The national debt is a stock, the budget balance is a flow. A budget deficit shows the increase in the national debt that occurs in one year

Post-monthly

Chapter 3 Free Trade

Interdependence:

Exports: goods produced domestically and sold abroad; to export means to sell domestically produced abroad

Imports: goods produced abroad and sold domestically; To import means to purchase goods produced in other countries

Absolute advantage: the ability to produce a good using fewer inputs than another producer

- The US has an absolute advantage in wheat: producing a ton of wheat uses 10 labor hours in the US vs 25 in Japan
- If each country has an absolute advantage in one good and specializes in that good, then both countries can gain from trade

Comparative advantage: the ability to produce a good at a lower opportunity cost than another producer

- Absolute advantage is not necessary for comparative advantage

Benefits of international trade/ free trade

- Increased Competition
- Lower prices
- Greater choices
- Access to larger markets
- Acquisition of resources
- Economies of Scale
- More efficient resource allocation

Chapter 9 Tariff and quota

P_W= the world price of a good; the price that prevails in world markets

P_D= domestic price without trade

If $P_D<P_W$:

- Country has comparative advantage in the good
- Under free trade, country exports the good

If $P_D> P_W$

- Country does not have comparative advantage
- Under free trade, country imports the good

The small economy assumptions

- A small economy is a price taker in world markets. Its actions have no effect on P_W
- Not always true—especially for the US—but simplifies the analysis without changing its lessons
- When a small economy engages in free trade, P_W is the only relevant price:
  - No seller would accept less than P_W. Since she could sell the good for P_W in world markets
  - No buyer would pay more than P_W, since he could buy the good for P_W in work markets

	$P_D < P_W$	$P_D > P_W$
direction of trade	exports	imports
consumer surplus	falls	rises
producer surplus	rises	falls
total surplus	rises	rises

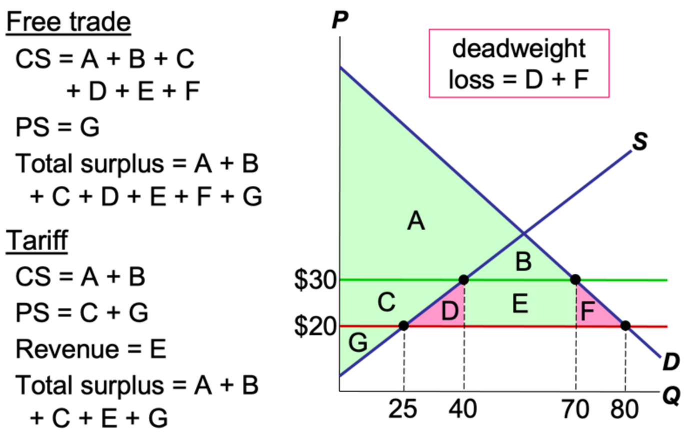
Whether a good is imported or exported,
trade creates winners and losers.
But the gains exceed the losses.

Other benefits of **International Trade**

- Consumers enjoy increased variety of goods.
- Producers sell to a larger market, may achieve lower costs by producing on a larger scale.
- Competition from abroad may reduce market power of domestic firms, which would increase total welfare.
- Trade enhances the flow of ideas, facilitates the spread of technology around the world.

Tariff: a tax on imports

- In general, the price facing domestic buyers & sellers equals ($\boldsymbol{P_W} + \boldsymbol{T}$).



An **import quota** is a quantitative limit on imports of a good.

- Mostly has the same effects as a tariff:
  - Raises price, reduces quantity of imports.
  - Reduces buyers' welfare.
  - increases sellers' welfare.
- A tariff creates revenue for the government. A quota creates profits for **the foreign producers** of the imported goods, who can sell them at higher price.
- Or, government could auction licenses to import to capture this profit as revenue. Usually it does not.

Export Subsidies – If a government subsidises domestic production this gives them an unfair advantage over competitors.

Administrative barriers – Making it more difficult to trade, e.g. imposing minimum environmental standards.

Arguments for Trade Protectionism

1. **The jobs argument**
- Trade destroys jobs in the industries that compete against imports.
- ***Economists' response:***
  - Total unemployment does not rise as imports rise, because job losses from imports are offset by job gains in export industries.
  - Even if *all* goods could be produced more cheaply abroad, the country need only have a *comparative* advantage to have a viable export industry and to gain from trade.
2. **The national security argument**
- An industry vital to national security should be protected from foreign competition, to prevent dependence on imports that could be disrupted during wartime.
- ***Economists' response:***
  Fine, as long as we base policy on true security needs.

But producers may exaggerate their own importance to national security to obtain protection from foreign competition.

3. **The infant-industry argument**

- A new industry argues for temporary protection until it is mature and can compete with foreign firms.
- ***Economists' response:***
  Difficult for govt to determine which industries will eventually be able to compete and whether benefits of establishing these industries exceed cost to consumers of restricting imports.
  Besides, if a firm will be profitable in the long run, it should be willing to incur temporary losses.

4. **The unfair-competition argument**

- Producers argue their competitors in another country have an unfair advantage *e.g.* due to govt subsidies.
- ***Economists' response:***
  Great! Then we can import extra-cheap products subsidized by the other country's taxpayers. The gains to our consumers will exceed the losses to our producers.

5. **The protection-as-bargaining-chip argument**

- Example: The U.S. can threaten to limit imports
  of French wine unless France lifts their quotas
  on American beef.
- ***Economists' response:***
  - Suppose France refuses. Then the U.S. must choose between two bad options:
  - A) Restrict imports from France, which reduces welfare in the U.S.
  - B) Don't restrict imports, which reduces U.S. credibility.

Trade Agreements

- A country can liberalize trade with unilateral reductions in trade restrictions and multilateral agreements with other nations
- Examples of trade agreements:
  - North American Free Trade Agreement (NAFTA), 1993
  - General Agreement on Tariffs and Trade (GATT), ongoing
  - World Trade Organization (WTO), est. 1995, enforces trade agreements, resolves disputes (164 members by 2016)
    - WTO's objective is to facilitate fair, transparent, and open global trade.
    - WTO's functions: Balances economic growth with dispute resolution and support for developing nations.
    - Plays a crucial role in maintaining global economic stability.

Chapter 31 Forex market and system

Variables that Influence Net Exports

- Consumers' preferences for foreign and domestic goods
- Prices of goods at home and abroad
- Incomes of consumers at home and abroad
- The exchange rates at which foreign currency trades for domestic currency
    - Ex: 1 usd=6.8 rmb→7.2rmb (usd appreciates→export decreases (US products more expensive)→import increases (one $ can buy more foreign goods)) NX decrease
    - China: depreciates, exports increase, imports decrease, NX increase
- Transportation costs
- Govt policies

Trade Surpluses and Deficits

NX measures the imbalance in a country's trade in goods and services

- Trade deficit: an excess of imports over exports
- Trade surplus: an excess of exports over imports
- Balanced trade: when exports=imports

The Flow of Capital

The flow of capital abroad takes two forms:

- **Foreign direct investment**:
  Domestic residents actively manage the foreign investment, *e.g.,* McDonalds opens a fast-food outlet in Moscow.
    - **Domestic government want because**: it creates job opportunities, promotes technological advancement
    - **Foreign government want because**: efficiency
- **Foreign portfolio investment**:
  Domestic residents purchase foreign stocks or bonds, supplying "loanable funds" to a foreign firm.

The Nominal Exchange rate

- **Nominal exchange rate**: the rate at which one country's currency trades for another
- We express all exchange rates as foreign currency per unit of domestic currency.

- **Appreciation** (or "strengthening"):
  an increase in the value of a currency
  as measured by the amount of foreign currency it can buy

- **Depreciation** (or "weakening"):
  a decrease in the value of a currency
  as measured by the amount of foreign currency it can buy

Demand Supply Diagrams

Why Demand curve is downward sloping: USD depreciates→us goods are cheaper→quantity demand for usd increases

Why the supply curve Is upward sploing: usd appreciates→uk products cheaper→Quantity supply of usd increases

Exchange Rate Systems

Fixed Exchange Rate System

This occurs when the government seeks to keep the value of a currency fixed against another currency. (Pegged to another currency)

- **Saudi Arabia (SAR)** – The Saudi riyal is pegged at 3.75 SAR per USD.
- **Hong Kong (HKD)** – Pegged at 7.8 HKD per USD since 1983.

Floating Exchange Rate System

When the government does not intervene in the foreign exchange market but allows market forces to determine the level of a currency.

- **United States (USD)** – The value of the dollar fluctuates based on forex markets.
- **Eurozone (EUR)** – The euro floats freely against other currencies.

A managed floating exchange rate (aka dirty float) is

- an exchange rate system that allows a nation's central bank to
  intervene occasionally in foreign exchange markets to change the direction of the currency's float and/or reduce the amount of currency volatility.

- **China (CNY)** – The People's Bank of China allows the yuan to trade within a controlled band.

- **Brazil (BRL)** – The Central Bank of Brazil sometimes intervenes in forex markets.

- **Singapore (SGD)** – The Monetary Authority of Singapore manages the SGD against a trade-weighted basket of currencies.